

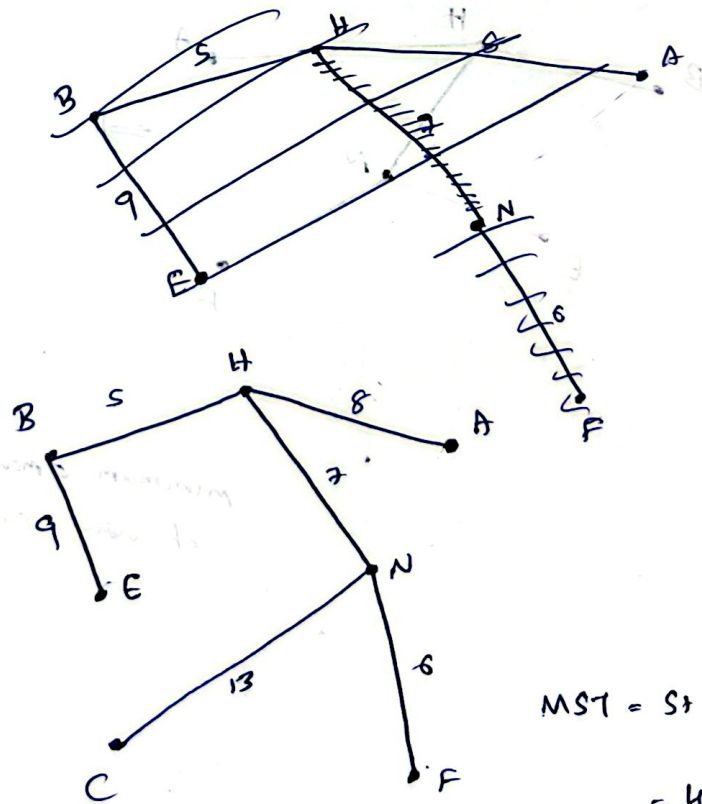
Registration No. 2025/CS/079

Index No. 25000799

(1). Tree - A connected graph with no cycles (loops)

- Spanning Tree - A subgraph of a tree that includes all the vertices of the original tree with minimum possible edges.
- Minimum Spanning Tree - A spanning tree for weighted graph where sum of weights of all its edges is minimized.
- Kruskal's Algorithm - Builds MST by sorting all the edges in the graph in ascending order of their weights and connecting them by ensuring no cycles are formed.
- Prim's Algorithm - Builds MST by starting from an arbitrary single vertex and continuously growing tree by adding the minimum weight edge while ensuring no cycles are formed.

- 5 B-H ✓
- 6 N-F ✓
- 7 H-N ✓
- 8 H-A ✓
- 9 B-E ✓
- 10 E-F X
- 10 N-A X
- 11 A-F X
- 12 E-H X
- 13 C-N ✓
- 14 B-C
- 14 C-G
- 15 G-N



$$\begin{aligned} \text{MST} &= 5 + 6 + 7 + 8 \\ &\quad + 9 + 13 \\ &= 48 \end{aligned}$$

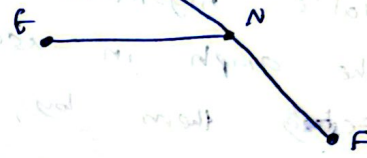
②

using Prim's Algorithm

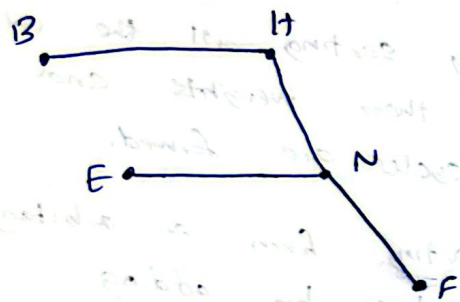
$\Rightarrow$



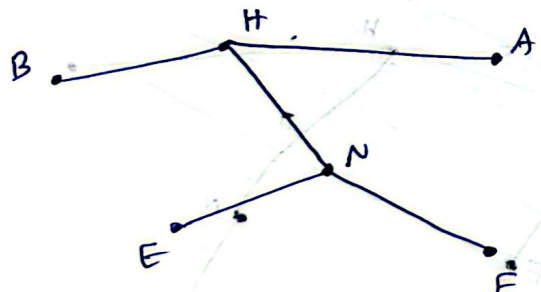
$\Downarrow$



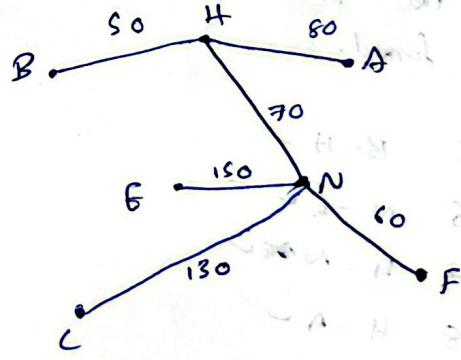
$\Leftarrow$



$\Downarrow$



$\Rightarrow$

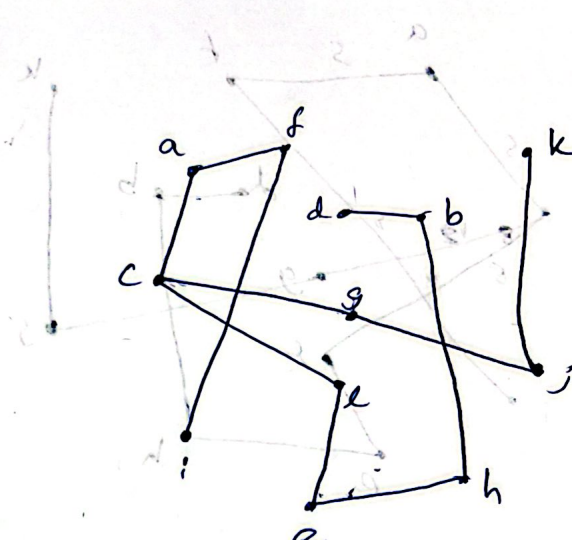


minimum amount of new cables $y = 50 + 60 + 70 + 80 + 130 + 150$
 $= 540m$

③

Graph A

5 - a-f ✓	10 c-i ✗	16 h-j
5 g-j ✓	12 c-g ✓	16 f-k
5 k-i ✓	12 b-h ✓	20 f-b
8 a-c ✓	13 g-l ✗	22 d-g
8 i-f ✓	14 a-l ✗	47 b-k
8 d-b ✓	15 e-l ✓	94 f-d
8 c-l ✓	15 l-h	
8 e-h ✓	16 i-e	

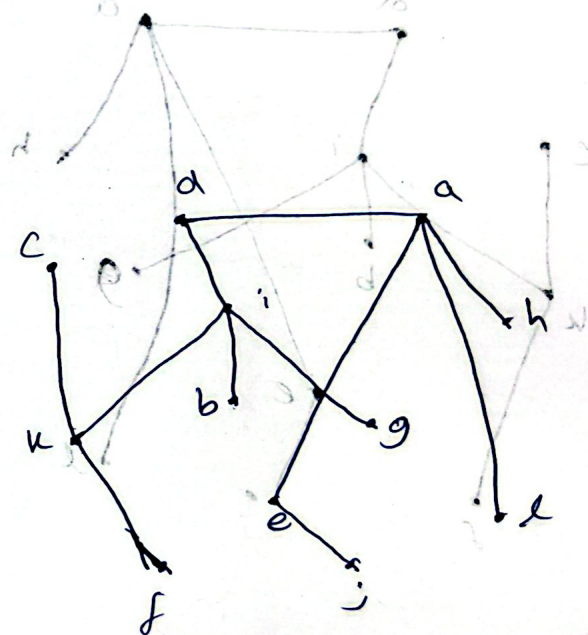


$$MST = 5 + 5 + 5 + 8 + 8 + 8 + 8 + 8 + 12 + 12 + 15$$

$$= 94 //$$

Graph B

1 c-k ✓	7 d-i ✓	20 a-cd
1 k-f ✓	10 b-g ✗	25 c-f
1 b-l ✓	10 k-g ✗	25 j-h
4 k-i ✓	10 a-h ✓	
4 b-f ✗	12 b-a ✗	
4 d-a ✓	15 a-l ✓	
5 i-g ✓	17 e-j ✓	
6 e-a ✓	18 f-l	



$$MST = 1 + 1 + 1 + 4 + 4 + 5 + 5 + 6 + 7 + 10 + 15 + 17$$

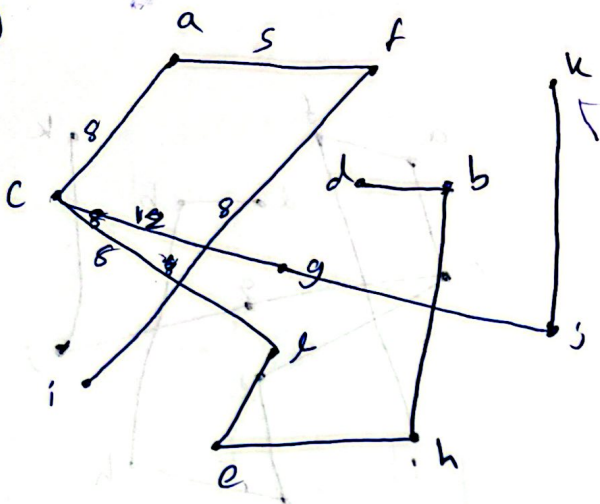
$$= 71 //$$



$$1 + 1 + 1 + 4 + 4 + 5 + 5 + 6 + 7 + 10 + 15 + 17 = 71 //$$

$$= 71 //$$

(k)

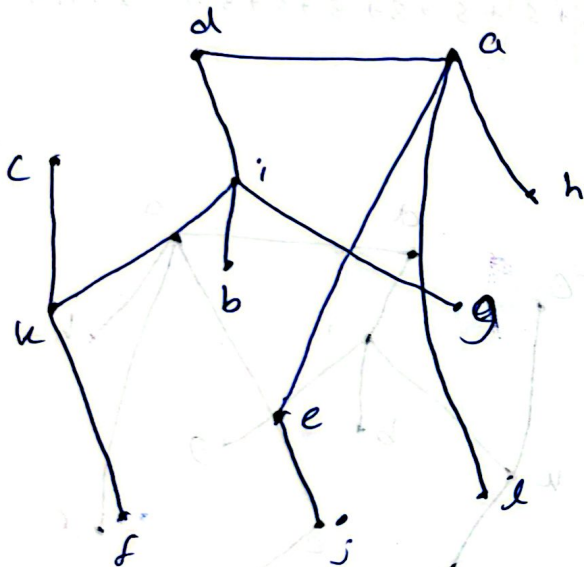


Graph A

$$MST = 94$$

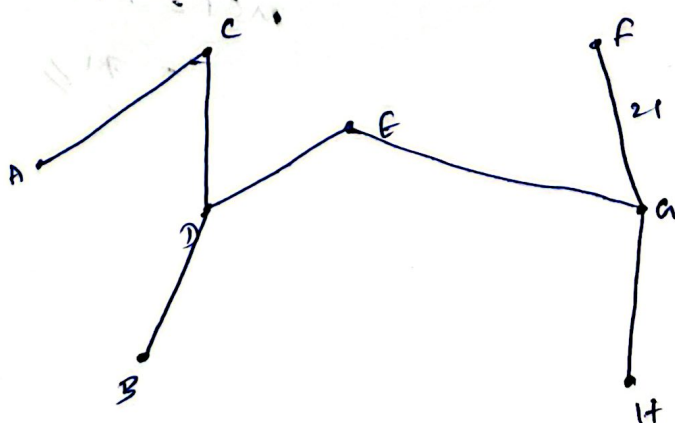
Graph B

$$MST = 71$$



(S)

- (A)
- | | | | |
|----|-------|----|-------|
| 18 | D-E ✓ | 31 | A-B ✗ |
| 21 | F-G ✓ | 33 | H-G ✓ |
| 22 | C-D ✓ | 34 | D-H |
| 24 | B-D ✓ | 38 | B-H |
| 24 | C-F ✗ | | |
| 26 | G-H ✓ | | |
| 28 | E-F ✗ | | |
| 29 | C-F ✗ | | |
| 30 | A-C ✓ | | |



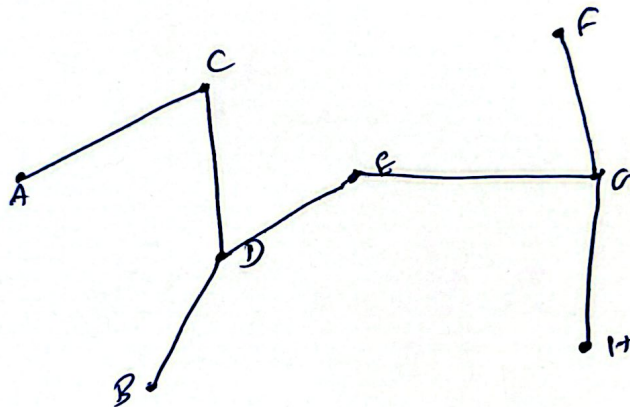
$$MST = 18 + 21 + 22 + 24 + 28 + 30 + 33 = 174 //$$

(B)

	A	B	C	D	E	F	G	H
A	-	31	30	-	-	-	-	-
B	31	-	-	24	-	-	-	38
C	30	-	-	22	24	29	-	-
D	-	24	22	-	18	-	-	34
E	-	-	24	18	-	28	26	-
F	-	-	29	-	28	-	21	-
G	-	-	-	-	26	21	-	33
H	-	38	-	34	-	-	33	-

(C)

- A → C
- C → D
- D → E
- D → B
- E → G
- G → F
- G → H



(D) $MST = 18 + 21 + 22 + 24 + 26 + 30 + 33$
 $= 174$

